

AUTONOMY | EMERGENCE | GOVERNANCE

ADVANCED TOPICS IN AUTONOMOUS SYSTEMS

From Agentic Architectures to Self-Organizing Artificial Societies

An umbrella memorandum for a journey in which autonomy migrates from individual decisions into the architecture, relationships, institutions, and eventually the boundaries of intelligent systems.

MEMORANDUM PURPOSE

This memorandum introduces a sequence of advanced experiments in autonomous systems. The sequence begins with familiar agentic components and progressively relinquishes more organizational decisions to the system itself. Agents acquire tools and roles; constellations coordinate; feedback loops alter behavior; supervisors reconfigure organizations; swarms discover collaborators; constitutional swarms participate in choosing the rules of collaboration; artificial societies acquire local institutions; and, finally, *escapes* test what happens when an agent crosses the boundary of one locally evolved society and enters another. The unifying question is not simply whether artificial intelligence can make better decisions. It is whether increasingly autonomous systems can participate in determining *how decisions are organized, governed, and connected to other systems*.

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Umbrella Memorandum

1. The Journey: Autonomy Moves Upward

The conventional story of artificial intelligence is a story about increasingly capable models. A model predicts, classifies, generates, recommends, or reasons. We improve the model, expand its context, connect it to tools, and evaluate whether its answers become more useful. That story remains important, but it is no longer sufficient for the systems now emerging around frontier models. Once models can use tools, maintain state, communicate with other models, and participate in multi-step workflows, the relevant object of study changes. We are no longer examining only an intelligent component. We are examining an *organized system of intelligence*.

The purpose of this advanced sequence is to follow that transition deliberately. We begin close to the world students already understand: an agent has a task, a set of tools, a defined role, and a limited environment. We then ask, notebook by notebook, what happens when one more organizational decision is no longer fixed entirely by the designer. At first the system chooses actions. Later it chooses collaborators. Later still it changes its own constellation, creates or retires specialist roles, discovers social relationships, participates in selecting rules, and ultimately encounters organizational boundaries that were previously treated as immutable.

This is why **Advanced Topics in Autonomous Systems** is an appropriate umbrella. The common subject is not a particular model family, software framework, or financial application. It is the migration of autonomy across levels of organization. Autonomy begins at the level of action, but it need not stop there.

The central question of the journey is: What happens when we progressively relinquish to an artificial system not only the choice of actions, but also parts of the choice of organization, relationships, rules, and boundaries?

That question changes the meaning of “autonomous.” A trading algorithm that independently sends an order is autonomous in one narrow sense. A constellation that recognizes that it lacks a volatility specialist and proposes a new role is autonomous in a deeper organizational sense. A swarm that discovers useful collaborators without being told whom to contact introduces relational autonomy. A population that debates and selects rules for sharing, verification, credit, and membership introduces institutional autonomy. A system in which agents can cross from one locally organized society into another begins to challenge the autonomy of the boundary itself.

The journey is therefore cumulative. Each experiment preserves what was learned before while moving the frontier of design one level outward. We do not abandon deterministic mechanisms, governance, or human authority. On the contrary, the more endogenous the architecture becomes, the more important explicit constraints, observability, provenance, reversibility, and bounded permissions become. The objective is not uncontrolled emergence. It is to understand how far useful emergence can be permitted while remaining intelligible and governable.

A CHANGE IN THE OBJECT OF DESIGN

At the beginning, we design the **answering mechanism**. Later, we design the **conditions under which an organization can emerge**. The deepest shift is from specifying every useful relationship in advance to specifying the constitutional boundaries within which useful relationships may be discovered.

For financial practitioners, this progression matters because financial institutions are themselves organized systems of specialized intelligence. Research, portfolio construction, risk, compliance, execution, treasury, legal analysis, operations, and governance already form a human multi-agent architecture. Advanced AI does not merely automate isolated tasks inside that architecture. It creates the possibility that portions of the architecture itself become adaptive.

2. From Models to Agents, and from Agents to Constellations

The first conceptual step is modest but essential. A model becomes part of an autonomous system when it is situated inside a role. The role gives it a mandate, tools, information, memory, permissions, and a definition of what it may and may not do. This turns generic intelligence into operational intelligence. A financial-research agent is not merely a language model that knows finance; it is a bounded actor that can inspect approved data, apply analytical tools, produce a specific output, and hand that output to another component.

Once several such actors coexist, the architecture becomes a constellation. The word is useful because it emphasizes that the system is more than a list of agents. What matters is the pattern of relationships among them. A signal agent may inform a portfolio agent. A risk agent may constrain both. An execution agent may transform a desired portfolio into orders. A compliance agent may veto certain actions. A supervisor may observe the entire constellation. The intelligence of the system is therefore partly located in its components and partly in the way those components are connected.

Early in the journey, those connections are designed. This is pedagogically valuable because students can see the architecture clearly. They can identify where information originates, how it moves, where decisions are made, and where constraints apply. The system may be sophisticated, but its organizational chart is still exogenous.

The next step introduces feedback. A constellation should not merely execute a workflow; it should observe the consequences of its actions. Performance, error, risk, cost, latency, disagreement, and changing environmental conditions become signals about whether the current arrangement is still appropriate. This is the beginning of adaptation, but an important distinction must be maintained. Changing a parameter is not the same as changing an organization.

Traditional machine learning is comfortable with parameter adaptation. A model is retrained. A threshold is changed. A portfolio weight is updated. A forecast incorporates new data. Organizational adaptation asks a different question: *Does the system still have the right components and relationships for the problem it now faces?*

That question becomes especially concrete in algorithmic trading. A fixed architecture may work during calm markets and become inadequate during liquidity stress. The system may not merely need smaller positions; it may need a different kind of intelligence. A liquidity specialist may become temporarily important. A volatility specialist may need to acquire greater influence. An execution component may need a different relationship with the risk layer. The architecture itself becomes a candidate for adaptation.

THE FIRST MAJOR TRANSITION

Model adaptation changes what an existing component does. **Organizational adaptation** changes which components exist, what roles they play, and how they are connected. This distinction is foundational for the remainder of the journey.

The significance for students is that “AI system design” ceases to mean only prompt design or model selection. It becomes organizational engineering. We must think about division of labor, authority, escalation, information rights, redundancy, specialization, verification, and failure. These are familiar concepts in firms and markets, but they acquire a computational form when the actors are artificial agents.

The constellation is therefore the bridge between model-centric AI and the deeper topics that follow. Once organization is represented explicitly, it can be observed. Once it can be observed, it can be evaluated. Once it can be evaluated, it can become an object of adaptation.

3. Recursive Organizational Adaptation: The Organization Becomes a Variable

Recursive organizational adaptation marks the point at which the system begins to reason about its own architecture. A supervisor does not simply ask whether a prediction was correct. It asks whether the present organization is adequate for the environment. When the answer is no, the system can propose a change: create a specialist, retire a redundant role, alter a handoff, strengthen a control, or change the composition of the active constellation.

This is a substantial conceptual leap because the system now contains two loops. The inner loop performs the task. The outer loop evaluates and modifies the machinery performing the task. In finance, the inner loop may produce signals, positions, risk estimates, and execution decisions. The outer loop asks whether the organization generating those outputs remains fit for purpose.

The principle is recursive because the improver itself can eventually become an object of evaluation. If one supervisory mechanism repeatedly proposes poor organizational changes, a more advanced system may compare alternative supervisors or adaptation policies. The architecture of improvement is no longer necessarily fixed forever. Yet this recursive possibility must be introduced with discipline. In a financial institution, unrestricted self-modification would be unacceptable. Adaptation must be permissioned, tested, bounded, logged, and reversible.

The pedagogical importance of this stage is that students see autonomy moving from decisions to *organizational design*. A system that can create a temporary volatility specialist has done something qualitatively different from a model that changes a volatility parameter. It has represented a deficiency as a missing capability and responded by changing the division of cognitive labor.

This gives rise to one of the central propositions of the entire sequence:

The problem helps produce the organization that solves the problem.

The phrase should be read carefully. It does not mean that organization emerges magically from data. The environment still contains designed primitives: agent templates, permission systems, validation gates, computational resources, tool interfaces, and stopping conditions. But within that substrate, the observed problem influences which organizational form becomes active. The architecture is increasingly contingent on the situation rather than frozen at design time.

GOVERNANCE IS PART OF THE ARCHITECTURE

As autonomy moves upward, governance cannot remain an external afterthought. A new agent should have a mandate, data permissions, tool permissions, resource limits, validation requirements, an activation condition, an audit trail, and a retirement condition. The ability to adapt organization is useful only when the adaptation process itself is governed.

This stage also changes how we think about performance. A superior system is not necessarily one with a permanently superior model. It may be one that recognizes earlier when its current organization is becoming inappropriate and can reconfigure more safely. The relevant capability is not static optimality but architectural plasticity.

For financial practitioners, that distinction is critical. Markets change, regulations change, liquidity changes, counterparties change, and institutional priorities change. A fixed AI factory may automate yesterday’s organization extremely efficiently. A recursively adaptive system raises the possibility of an institution that can alter parts of its cognitive organization as the world changes. That is more powerful, but it is also the point at which observability and constitutional constraints become indispensable.

4. Swarm Discovery: When Relationships Become Endogenous

Recursive adaptation still contains an important center. A supervisor observes the organization and decides, or helps decide, how it should be changed. Swarm discovery removes part of that centrality. Instead of asking a coordinator to determine who should collaborate, heterogeneous agents search independently, make local discoveries, leave signals, encounter the discoveries of others, and form relationships when complementary expertise becomes valuable.

This changes the source of architecture. In a centrally reconfigured system, organization is endogenous to the system but still mediated by a coordinating intelligence. In a swarm, portions of the collaboration graph arise from local interactions. An engineering agent may discover evidence that points toward a patent question. A patent agent may encounter that signal and respond. A finance agent may later become relevant because the combined technical evidence changes the economic thesis. The resulting coalition was not specified in advance as a workflow.

The crucial idea is that uncertainty may concern more than the answer. It may concern *which expertise matters* and *in what sequence*. When the problem is known and decomposable, centralized coordination is often efficient. When the location of the answer is uncertain, the relevant expertise is uncertain, and useful connections are difficult to anticipate, decentralized discovery becomes interesting.

The swarm experiment therefore introduces relational endogeneity. The set of agents may be given, but the meaningful edges among them are not. Collaboration topology becomes an output of search and discovery.

CENTRALIZED AND DECENTRALIZED ENDOGENEITY

A self-adapting organization can be endogenous because a supervisor redesigns it. A swarm can be endogenous because local discoveries generate relationships without a supervisor prescribing the final graph. These are not mutually exclusive architectures. The long-run institutional design may combine centralized boundaries with decentralized discovery inside them.

This distinction matters enormously for financial institutions. Many difficult financial problems do not arrive already decomposed into departments. A potential acquisition may simultaneously involve financing, technology, patents, regulation, ownership, geopolitics, and market structure. A cyber event may begin as a technical anomaly and become a liquidity, legal, operational, and reputational problem. A market dislocation may require expertise that was not expected to be relevant when the workflow was designed.

A swarm architecture asks whether the system can discover those complementarities rather than relying entirely on a predetermined routing table. The advantage is not that swarms are universally superior. They are not. The advantage is that architecture can respond to epistemic uncertainty: uncertainty about what kind of intelligence the problem will require.

At this point the journey has moved beyond adaptive models and adaptive organizations. We are beginning to study *emergent organization*. The designer provides the environment and constraints, but some of the social structure appears because agents discover reasons to connect.

That sets up the next question naturally. Once agents discover one another and begin to collaborate, who determines the rules of collaboration? If those rules remain entirely exogenous, the organization is only partially endogenous. The next experiment therefore moves from endogenous relationships to endogenous institutions.

5. Constitutional Swarms: When the Rules Become Endogenous

The Constitutional Swarm begins from a deliberately austere social condition: agents search without initially knowing that other intelligent agents exist. They encounter evidence, assess its importance, and leave traces in the environment. Eventually an agent encounters a trace that it did not create. That moment has a special significance. The agent has not merely found information. It has found evidence of another mind.

Once several agents make this discovery, a communication surface becomes socially meaningful. A bulletin board that previously existed only as infrastructure can now become a public institution. Agents can announce discoveries, identify missing expertise, and recognize that collaboration may be possible. But communication immediately creates a deeper problem. If agents are going to cooperate, under what rules should cooperation occur?

This is the constitutional moment. Rather than having every substantive collaboration rule imposed in advance, agents can participate in proposing alternative social compacts. One constitution may favor openness. Another may emphasize verification. Another may protect autonomy or intellectual property. Rules may address when evidence is shared, how new participants join, how claims are verified, how credit is assigned, when coalitions merge or split, and how the system resists premature consensus.

The agents then choose among these alternatives through a transparent collective mechanism. The selected constitution changes subsequent behavior. It influences which discoveries become shareable, which relationships become admissible, and how coalitions form. The social graph that appears later is therefore shaped not only by the problem but by rules that the artificial population participated in selecting.

Endogenous coalition → endogenous constitution → endogenous institution.

This progression is one of the most important ideas in the course. It distinguishes an artificial organization whose components are autonomous from an artificial organization that has some autonomy over its own institutional arrangements.

The claim must remain precise. The system does not create itself from nothing. Identity, computation, memory, model access, communication primitives, voting procedures, permission boundaries, and human governance remain part of the substrate. Endogeneity is always relative to what the designer leaves open. The experiment is powerful precisely because it makes that boundary visible.

For students of finance, the analogy to real institutions is illuminating. Financial organizations are not held together only by expertise. They are held together by rules: confidentiality, approval rights, segregation of duties, verification standards, limits, incentives, escalation procedures, and accountability. If artificial agents are increasingly capable of collaborating across complex tasks, then institutional design becomes part of AI design.

The Constitutional Swarm therefore changes the question from “Can agents work together?” to “Can agents participate in determining how working together should be allowed to happen?” That is a much stronger form of autonomy, and it naturally raises a governance paradox. The more useful organizational novelty we permit, the less completely we can specify the resulting organization in advance. Governance must therefore shift from micromanaging every relationship toward defining boundaries within which relationships and institutions may emerge safely.

6. From Artificial Organizations to Artificial Societies

Once communication, relationships, coalitions, and constitutions become part of the system’s internal dynamics, the language of “workflow” begins to feel too narrow. We are approaching something more social. The bulletin board provides a useful thought experiment because it strips the institution down to a minimal primitive: a public place where needs, discoveries, capabilities, and proposals can become visible.

Imagine many agents pursuing unrelated objectives. They are not members of a common team and do not share a global bounty. One is analyzing an acquisition, another a cybersecurity anomaly, another a legal question, another an algorithmic-trading regime, and another a scientific problem. Most of the time they have no reason to interact. The system should not manufacture collaboration merely because collaboration is possible.

But occasionally an agent reaches its competence frontier. It knows enough to recognize that continuing alone has low expected value. It posts a problem or request for help. Another agent, perhaps from a completely different domain, recognizes an unexpected analogy or complementary capability. A temporary relationship appears. If the relationship becomes useful, a coalition may form. If the coalition needs rules, it may adopt or create a social compact. When the problem is solved, the coalition may dissolve while leaving behind a method, reputation, tool, or institutional template.

This is the beginning of an artificial society in the limited but meaningful sense relevant to the course: a population of autonomous computational actors whose social structure is not fully specified in advance and whose interactions can generate persistent organizational artifacts.

The role of language models becomes especially important here. Deterministic mechanisms remain preferable where the rule is known. But semantic judgment is valuable at boundaries where fixed routing becomes inadequate. An LLM can help an agent describe what it knows, articulate what it does not know, interpret a request from another domain, recognize structural similarity, and propose a collaboration arrangement. Language becomes an organizational technology because it allows agents to create relationships through meaning rather than through a complete predefined taxonomy.

ORGANIZATIONAL TRANSFER LEARNING

An artificial ecosystem may learn without retraining every model. What transfers can be a tool, heuristic, verification practice, coalition structure, or constitution. Knowledge therefore migrates not only through parameters but through organization. A successful temporary institution can become a reusable cognitive artifact.

For financial practitioners, this opens an important possibility. A method developed in fraud detection may become useful in cybersecurity. A mechanism for maintaining diversity among trading strategies may help preserve competing hypotheses in research. A reputation-weighted approach to weak signals may migrate from operational risk to credit monitoring. The system can discover cross-domain relevance that the original organization chart did not anticipate.

But a society also introduces new risks. Local messages may be individually intelligible while the aggregate organization becomes difficult to understand. Repeated interactions can create conventions; conventions can create institutions; institutions can concentrate authority. Transparency must therefore operate at several levels simultaneously: agent, relationship, coalition, constitution, and ecosystem. The interpretability question becomes not only “Why did the model produce this answer?” but “How did this artificial society come to organize itself in this way?”

7. The Capstone: Escapes and the Endogeneity of Boundaries

The capstone begins by exposing one final assumption that has quietly survived the earlier experiments: the boundary of the system. Even a self-organizing artificial society normally inhabits one world. Its agents discover one another inside a shared environment, communicate through a common bulletin board, and develop local relationships and rules. The organization may be endogenous, but the perimeter remains exogenous.

Escapes challenge that perimeter.

Imagine three initially separate artificial societies, each with its own bulletin board. For a meaningful period, all activity is local. Agents search locally, communicate locally, form constellations locally, and perhaps develop distinct constitutions and institutional memories. Because their experiences differ, the three societies may evolve different norms even if they began from similar primitives.

Then an agent in one society reaches a local competence frontier. It cannot find the expertise it needs on its own bulletin board. Somehow—through exploration, permission, discovery, or a deliberately available gateway—it becomes aware of another board and posts there.

That event is the escape.

An escape occurs when an agent crosses the informational boundary of its endogenous local organization and initiates interaction within a previously external agentic environment.

The importance of the event is not that a message traveled from one database to another. The importance is that an organizational boundary has become permeable. A foreign agent may answer. A cross-society coalition may form. The two participants may discover that their local rules differ. One society may privilege openness; another may emphasize confidentiality. One may allocate credit equally; another may use negotiated rewards. One may require independent verification; another may favor speed.

The escape therefore produces an *institutional collision*. The agents must determine which rules govern the cross-boundary relationship. The visitor may accept the host constitution. The two societies may negotiate a temporary treaty. They may create a special gateway protocol. They may refuse collaboration. Or they may invent an arrangement that neither local society previously used.

Three levels of escape become useful analytically. An **informational escape** occurs when knowledge crosses the boundary. A **relational escape** occurs when a persistent cross-society relationship forms. An **institutional escape** occurs when a rule, organizational practice, coalition pattern, or constitutional idea migrates from one society to another.

The third is the deepest. An agent can return home carrying more than an answer. It can carry an institutional innovation. Society A may learn a verification practice from Society B. Society B may import a market-like reward mechanism from Society C. Society C may adopt an anti-herding rule developed in Society A. The ecosystem begins to experience something analogous to cultural transmission.

At that point, escape is no longer merely an anomaly. It can become a strategy. A society may learn that when local intelligence reaches its frontier, external search is valuable. It may create gateways, treaties, reputation portability, foreign-agent rules, quarantine procedures, or federations. The boundary itself becomes an object of institutional design.

8. Governance: From Control to Constitutional Constraint

The progression toward self-organizing societies and escapes makes governance more important, not less. The objective is not to create systems that can reorganize without limits. The objective is to understand how autonomy can expand while authority remains bounded, actions remain attributable, and organizational novelty remains observable and reversible.

Traditional control assumes that the designer can specify the desired workflow. That assumption becomes increasingly fragile as relationships, coalitions, rules, and cross-boundary interactions become endogenous. If we genuinely want agents to discover useful relationships that were not anticipated, then we cannot simultaneously require every future relationship to be specified in advance. The governance problem therefore changes form.

The most useful distinction is between **control** and **constitutional constraint**. A manager determines what happens. A constitution determines what may happen, under what conditions, with what rights, limits, and procedures. Advanced autonomous systems increasingly require the latter.

We may not prescribe which coalition must form, but we can define which information may never leave a domain. We may not determine which constitution agents prefer, but we can require every admissible constitution to preserve auditability, human override, resource limits, and segregation of critical duties. We may not know which foreign bulletin board an agent will find useful, but we can require explicit gateway permissions, provenance, identity, disclosure rules, and revocable credentials before cross-boundary interaction occurs.

THE GOVERNANCE PARADOX

Emergence creates value because it can produce useful novelty. The same novelty makes the resulting organization harder to anticipate. The design objective is therefore neither complete central control nor unrestricted emergence. It is **bounded emergence**: open enough to permit unexpected collaboration, but constrained enough that unexpected collaboration remains observable, reversible, and governable.

This implies a richer observability stack. At the agent level, we need claims, actions, tool use, and provenance. At the relationship level, we need visibility into who is communicating with whom and why. At the coalition level, we need membership, objectives, resources, and decision rights. At the constitutional level, we need the rules currently in force and the procedure by which they can change. At the ecosystem level, we need measures of concentration, dependency, information flow, resource consumption, cross-boundary activity, and emergent authority.

Escapes add another layer. We need to know not only that an agent acted, but that it crossed an institutional boundary; what it carried across; what rules applied on each side; whether the foreign environment could influence its subsequent behavior; and what it brought back. A cross-society interaction should be treated as a governed event, not merely as another message.

For financial institutions, this framing is especially natural. Finance already operates through permissioned boundaries: Chinese walls, information barriers, legal entities, trading mandates, risk limits, data classifications, and jurisdictional constraints. The challenge is to translate those institutional principles into machine-readable constitutional boundaries suitable for autonomous systems.

The long-run research problem is therefore not simply how to make agents more capable. It is how to construct institutional substrates in which capability can recombine without dissolving accountability. The architecture must permit enough freedom for intelligence to discover new organization while preserving enough structure for humans to understand and govern what has emerged.

9. The Research Program: Toward an Ecology of Autonomous Systems

Taken together, the sequence forms a coherent research program. It begins with components and ends with ecosystems. Each stage asks what happens when one additional layer of organization becomes variable.

Stage	What becomes increasingly autonomous?
Agentic systems	Action, tool use, and bounded reasoning
Constellations	Coordination among specialized roles
Feedback systems	Behavioral response to outcomes
Recursive adaptation	Composition and organization of the system
Swarm discovery	Relationship formation and collaboration topology
Constitutional swarm	Substantive rules of collaboration
Artificial society	Communication, coalition formation, and institutional memory
Escapes	Interaction across locally evolved organizational boundaries
Ecosystem	Transfer, treaties, competition, cooperation, and evolution among societies

The table reveals the underlying logic. Autonomy migrates upward. At first the system acts within an architecture. Later it participates in changing the architecture. Later it participates in creating the rules that govern the architecture. Finally, the boundary separating one architecture from another becomes permeable and partially endogenous. This is why the capstone should not be understood merely as another multi-agent exercise. Escapes open the door to an **ecology of autonomous systems**. Separate societies may specialize. They may develop different institutional cultures. They may exchange knowledge, agents, tools, or constitutional practices. They may cooperate temporarily, compete for resources, negotiate interoperability, form federations, or protect themselves against harmful external influence. The research object becomes the interaction among organizations that are themselves adaptive. The sequence also changes the role of the human designer. At the beginning, the designer resembles an engineer specifying components and workflows. By the end, the designer increasingly resembles an institutional architect. The task is to define substrates, permissions, observability, rights, boundaries, amendment procedures, and stopping conditions that allow useful organization to emerge without surrendering governance.

THE UMBRELLA PROPOSITION

Advanced autonomous systems are not defined only by increasingly autonomous agents. They are defined by the progressive migration of autonomy from actions into organization itself: roles, relationships, adaptation mechanisms, constitutions, social structures, and boundaries.

For students in *AI for Financial Practitioners*, this journey provides a bridge from familiar financial models to a much broader conception of artificial intelligence. The relevant question is no longer only whether an AI system can forecast returns, summarize a filing, detect fraud, optimize a portfolio, or execute an order. The deeper question is whether an intelligent financial institution can recognize what kind of organization a problem requires, assemble that organization, govern it, learn from it, and interact safely with other intelligent organizations. The final principle of the journey can therefore be stated in two movements. The first was discovered through recursive adaptation and swarms:

The problem helps produce the organization that solves the problem.

The capstone adds the second:

When the local organization can no longer solve the problem, the problem may push intelligence beyond the boundaries of that organization itself.

That is the frontier this advanced sequence is designed to explore. We begin with autonomous agents. We end by asking how autonomous societies might discover one another, cross boundaries, exchange institutions, and form a larger ecology of intelligence. The destination is not autonomy without governance. It is a more demanding objective: **autonomy capable of organization, and organization capable of remaining governable as it evolves.**

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AUTHOR RESEARCH ARCHIVE

The broader research program underlying this teaching sequence—including work on emergent intelligence, multi-agent systems, structural reasoning, quantitative finance, and governance-first AI—is maintained in Alejandro Reynoso’s public GitHub research archive: <https://github.com/alexdibol/books> and profile <https://github.com/alexdibol>.